Assignment Choice 1 – Taxi Demand Forecasting

Description:

You are a newly recruited data science team member for Gett, New York City branch, and the company is undergoing a challenging period since it only recently entered the ride-sharing market in NYC and is still "learning" the market and how it functions in order to tailor its business operations effectively to the people and userbase in NYC.

In your first week at work, your manager called you in for an urgent meeting with the Fleet Operations team head. Once you went into the meeting room with them and sat down, your manager said "Hope you took your DSMA course during your Bachelor's degree at BGU, you're going to need to use everything that you learned to get us through this difficult period" and handed you a document that lists the business question which the team urgently needs investigated. It reads as follows:

Does every location in New York City have the same amount of "demand" for taxis? We need to know this so that we understand the optimal use of our resources in determining which parts of the city have to be covered by how many taxis at specific times and days in order to position our vehicles and drivers at these areas effectively.

a. **Input** – Date and time of request to check for , PULocationID
b. **Output** (target variable) – "Demand" (Number of Taxis needed at the given locationID at a specific time on a specific day.)
c. **The challenge - This is a data aggregation problem**. You cannot use the dataset "as-is" (in the *trip_duration* regression that you did in the course, you used the dataset "as-is" – since the *trip_duration* target variable could be calculated for every trip (row) in the dataset. But here, you will need to perform "grouping" of the dataset and transform it suitably before you begin using it) Use the following hints for forming your target variable –
    i. Your initial dataset (after cleaning and validation) will probably look as follows:

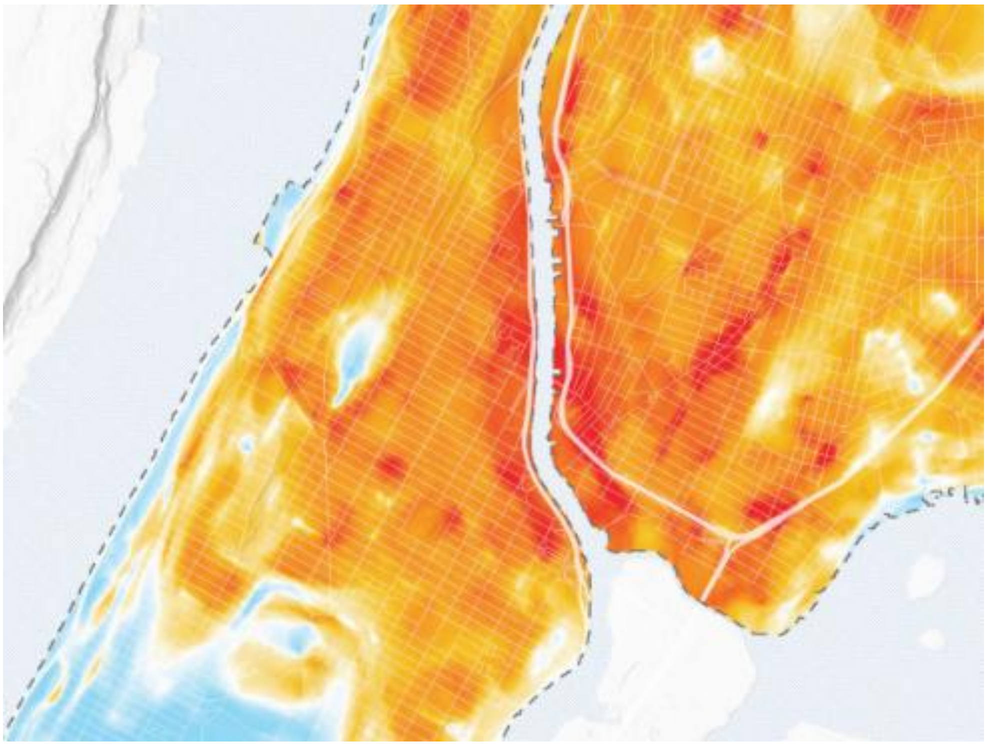
	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance	PULocationID	DOLocationID
0	2024-01-01 00:57:55	2024-01-01 01:17:43	1.0	1.72	186	79
1	2024-01-01 00:03:00	2024-01-01 00:09:36	1.0	1.80	140	236
2	2024-01-01 00:17:06	2024-01-01 00:35:01	1.0	4.70	236	79
3	2024-01-01 00:36:38	2024-01-01 00:44:56	1.0	1.40	79	211
4	2024-01-01 00:46:51	2024-01-01 00:52:57	1.0	0.80	211	148

    ii. But your final dataset before you begin the task of feature engineering and modelling should look something like this:

	pickup_hour	PULocationID	demand	hour	day_of_week	month
0	2023-12-31 23:00:00	68	1	23	6	12
1	2023-12-31 23:00:00	90	1	23	6	12
2	2023-12-31 23:00:00	138	1	23	6	12
3	2023-12-31 23:00:00	144	1	23	6	12
4	2023-12-31 23:00:00	161	1	23	6	12
...	...	...	...	...	...	...
77481	2024-01-31 23:00:00	263	53	23	2	1
77482	2024-01-31 23:00:00	264	4	23	2	1
77483	2024-02-01 00:00:00	138	1	0	3	2
77484	2024-02-01 00:00:00	161	1	0	3	2
77485	2024-02-01 00:00:00	186	1	0	3	2

(**Be careful** – most grouping functions will leave out hours where there was no demand (i.e., no trips) from the dataset – make sure you capture those too with a demand value of 0.)

d. **Build a simple Streamlit UI** that takes as input the time of the day from the user and outputs on screen a map of new York city with a heatmap overlay showing which parts of the city have the highest demand. (the intuition for this is that you use the ML model you trained to forecast "demand" for all the LocationIDs in New York for the given user input time and visualize it on a map plot after you have as output from the model the "demand" values for all the LocationIDs) Something like this:



e. **Data use constraints**: You may use the data over 2024 and 2025 as you wish, but reserve 2026 data specifically for evaluating and demonstrating how your model performs. (Do not include 2026 data in your training set.)

After staring at this document with beads of sweat forming on your forehead, you turn to leave the meeting room, when the Fleet Operations team head laughs and remarks "Thankfully, the municipality of New York has open sourced its historical data on traditional taxi rides in the city. There should be more than enough data in those data dumps that we can analyse to answer these questions." You run quickly to your laptop and run a search online to find these datasets and discover the following:

Dataset:

NYC – TLC (data from years 2024 and 2025, and JAN + FEB of 2026) – can be downloaded from here - https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Use only the "**Yellow Taxi Trip Records**" datasets from **2024**, **2025** and the **first two months of 2026**.

Data Dictionary:

Can be found here - https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf

And the Taxi Zone Lookup Table can be found here:

https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv

Use of AI:

Permitted, but please put in a little effort to understand the code that is generated (and to check if it is correct / useful for what you wish you achieve) before you use it in your codebase.

Evaluation Criteria:

1. **Completeness (20%)** – the extent to which the machine learning pipeline was built for answering this business question.
2. **Effectiveness of EDA (20%)** – How well you explored the data to find the initial insights in the data that allowed you to work effectively with the data.
3. **Innovation (20%)** – be creative with your feature engineering and model development
4. **Modularity of the Code (20%)** – Make the codebase that you build for this assignment modular and readable. One of the goals of this course is to highlight how modularity of the codebase makes it much easier to maintain your codebase and update it (especially when working in teams).
5. **Performance (10%)** – how well your model(s) work over the test set (you score 10/10 here if you beat the lecturer's model).
6. **Use of Experiment Tracking (10%) –** In the course, we use Weights & Biases (wandb) for the purpose of experiment tracking. I strongly encourage using experiment tracking when running your experiments (for the purpose of hyperparameter tuning using sweeps, for running evaluations of your models before and after you make changes to your features in order to see what works, and to also work with model and data versioning (as we will see in Lecture 4))

Assignment Submission format:

You are expected to submit 3 files:

a. **A python notebook** that you developed for performing the EDA that the lecturer can run over the training data (2024 + 2025 combined)

b. **A zip folder containing the entire codebase** you developed for the machine learning pipeline which the lecturer can execute that will run everything needed for the pipeline to work from start to finish. **Training data** – 2024 + 2025, **Test Data** – Jan + Feb of 2026
(include a text file containing instructions to execute the code if you feel it's needed)

c. **A PDF report** summarizing the following:
   a. Your key findings from the EDA connecting it to the business problem that you are trying to solve for. (e.g., It's important to keep most of our available drivers in LocationIDs XYZ specifically on Sundays because these are the only locations that are super active on Sundays)
   b. A short explanation of the final selected set of features that you engineered
   c. A short explanation of how you performed model selection and hyperparameter tuning
   d. A short explanation of the model improvement loops that you did (error analysis of your model + drift detection and mitigation)
   e. A link to your experiment tracking page on wandb (within the same team that you have joined for the course. You can open a new project for yourself.)